



BANK STATEMENT EXTRACTION & AUDITING USING AI

Course: Deep learning
Degree: AIM.

Agenda

1.Introduction

- Problem statement
- Problem scope

1.Business case

2.Technical

- Tech stack (Donut, PDF plumber, LLM)
- Pipeline outline
- explanation : Extraction > LLM > pipeline comparison
- Dashboard
- Future improvements : Fine tuning the DONUT, make it work of any document.

1.Conclusion

- Summary
- Key takeaways

INTRODUCTION

01

Automated Data Extraction

Streamlining the retrieval of transaction data from diverse bank statement formats.

02

Digital & Scanned PDF Support

Capability to process both native digital PDFs and scanned image-based documents.

03

AI Driven Validation

Leveraging artificial intelligence for robust data validation and enhanced accuracy.



The Challenge: Manual Document Processing



PDFs as Primary Format

Bank statements are overwhelmingly received in various PDF formats, often unstructured.

Time-Consuming Manual Work

Extraction of data from these documents is a labor-intensive and slow process.

High Risk of Human Error

Manual entry significantly increases the probability of costly mistakes and inconsistencies.

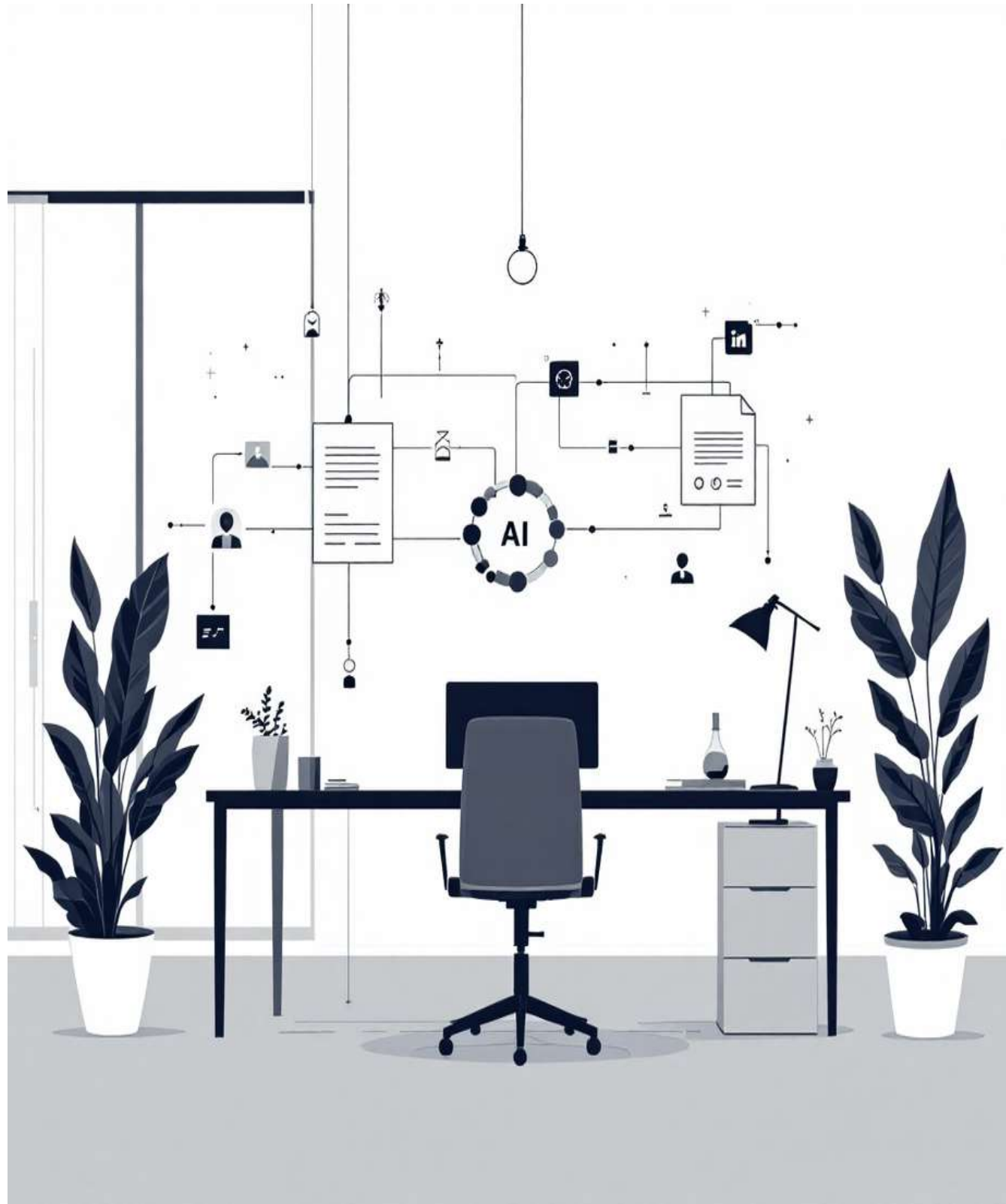
From Chaos to Clarity: The Problem We're Solving

Organizations frequently grapple with bank statements in PDF format, which are inherently not analysis-ready. This necessitates manual data extraction—a process notorious for being slow, error-prone, and highly inefficient. The complexity further escalates when dealing with scanned images of statements, where optical character recognition (OCR) challenges make digital processing incredibly difficult.

"Our goal: transform static, inaccessible data into dynamic, actionable insights."



Our Vision: Automating the Unautomatable



Our core motivation was to fully automate the entire bank statement processing workflow. We envisioned a comprehensive solution capable of handling both digital and scanned PDF formats, drastically minimizing errors, and converting complex, unstructured documents into clean, structured data ready for analysis and storage.

Project Scope: What We're Building

In Scope

- Bank statement PDFs (digital & scanned)
- Automated transaction data extraction
- Output in structured JSON format
- Comparative accuracy validation



Not in Scope

- Live bank API integrations
- Real-time transaction feeds
- Advanced financial forecasting
- Account reconciliation features



NEXT-GENERATION BANK STATEMENT PROCESSING

- In today's fast-paced business environment, automating bank statement processing is critical for ensuring financial accuracy, regulatory compliance, and operational agility.
- This presentation outlines key performance indicators (KPIs) used to evaluate the effectiveness of bank statement automation initiatives.
- These KPIs focus on driving measurable improvements in processing efficiency, data accuracy, compliance readiness, and cost optimization.



DRIVING OPERATIONAL EXCELLENCE: EFFICIENCY KPI'S



1

Processing Time per Receipt :

Aim for less than 15–30 seconds per statement, significantly accelerating financial operations and reducing reconciliation delays.

2

Throughput

Monitor the number of bank statement pages or statements processed per hour/day to ensure consistent, high-volume processing across multiple accounts and institutions.

3

Automation Rate

Target an 80–95% automation rate, minimizing manual intervention in transaction extraction, balance validation, and statement classification.

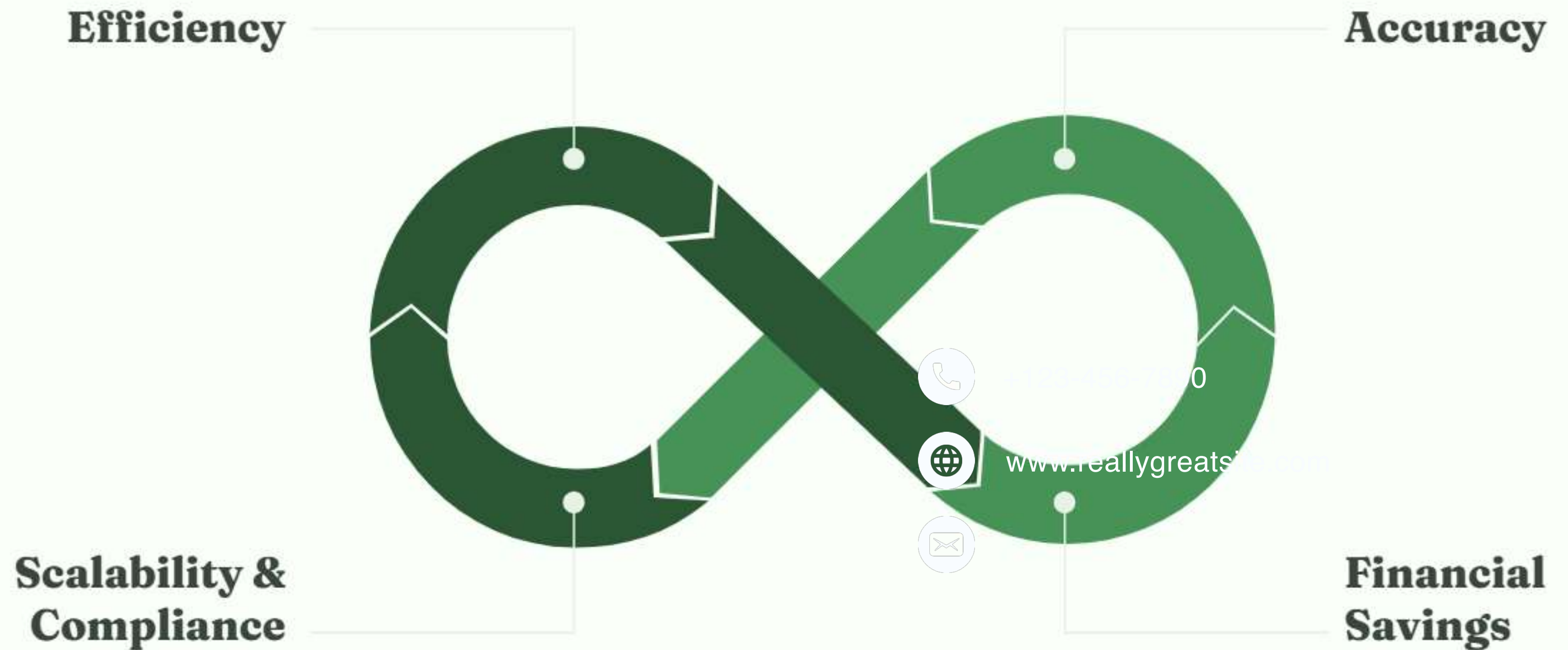
4

Human Touchpoints per Receipt

Strive for minimal to zero-touch processing, enabling finance teams to focus on exception handling, audits, and higher-value financial analysis.



THE INTEGRATED IMPACT OF AUTOMATION



FINANCIAL KPIS



Cost per Receipt

Target a 60–80% reduction in processing cost per bank statement through automation, eliminating manual data entry and reconciliation effort.



Operational Cost Savings

Track measurable monthly and yearly cost savings driven by reduced manual labor, faster statement reconciliation, and streamlined financial workflows.



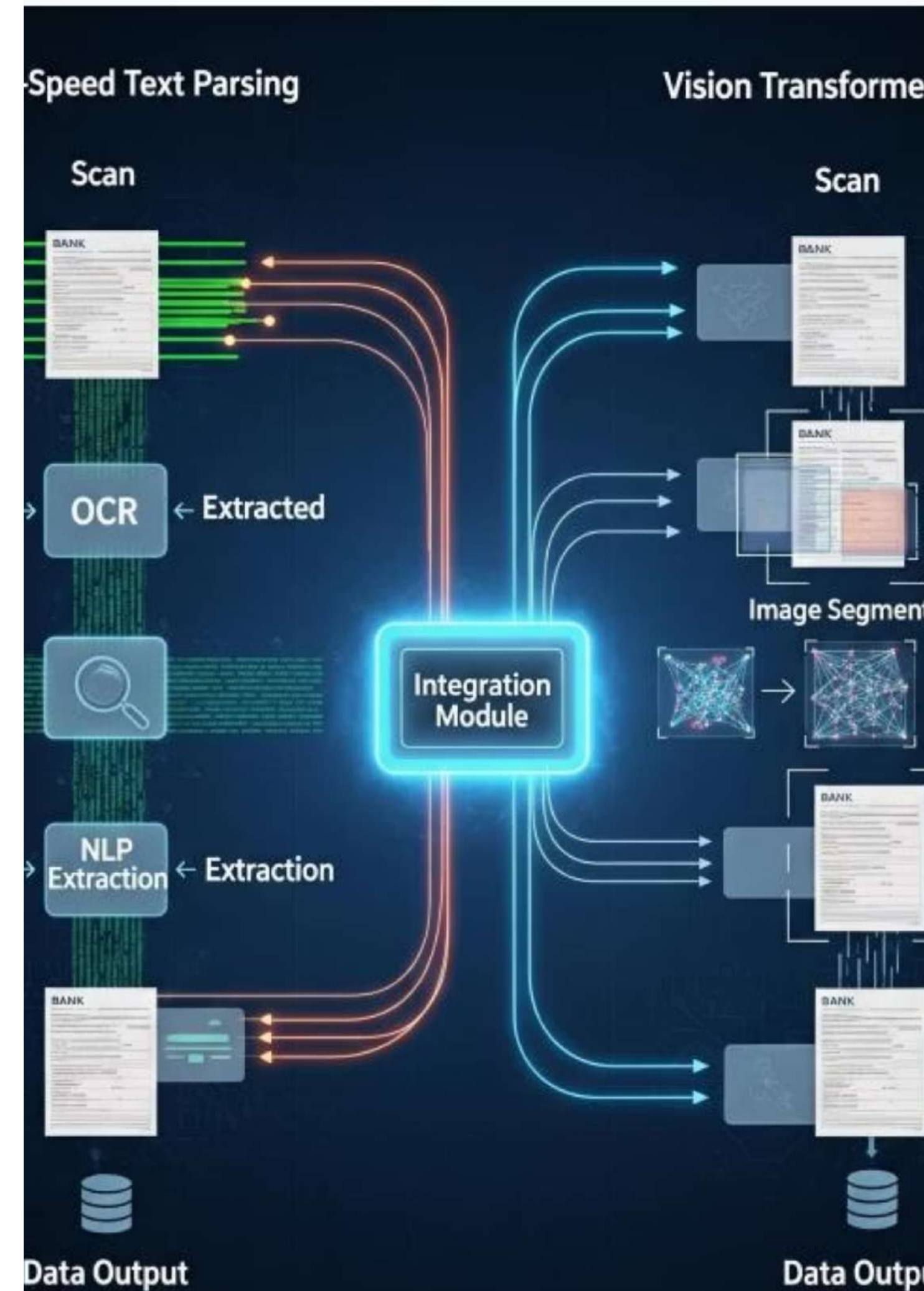
Return on Investment (ROI)

Aim for a rapid payback period of 3–6 months on bank statement automation investments, especially in high-volume finance operations.



Pipeline Bank and Receipts Data Extraction

A dual-pipeline system extracts data from bank documents using two powerful approaches: High-Speed Text Parsing and Vision Transformers.



Two Paths to Data Extraction



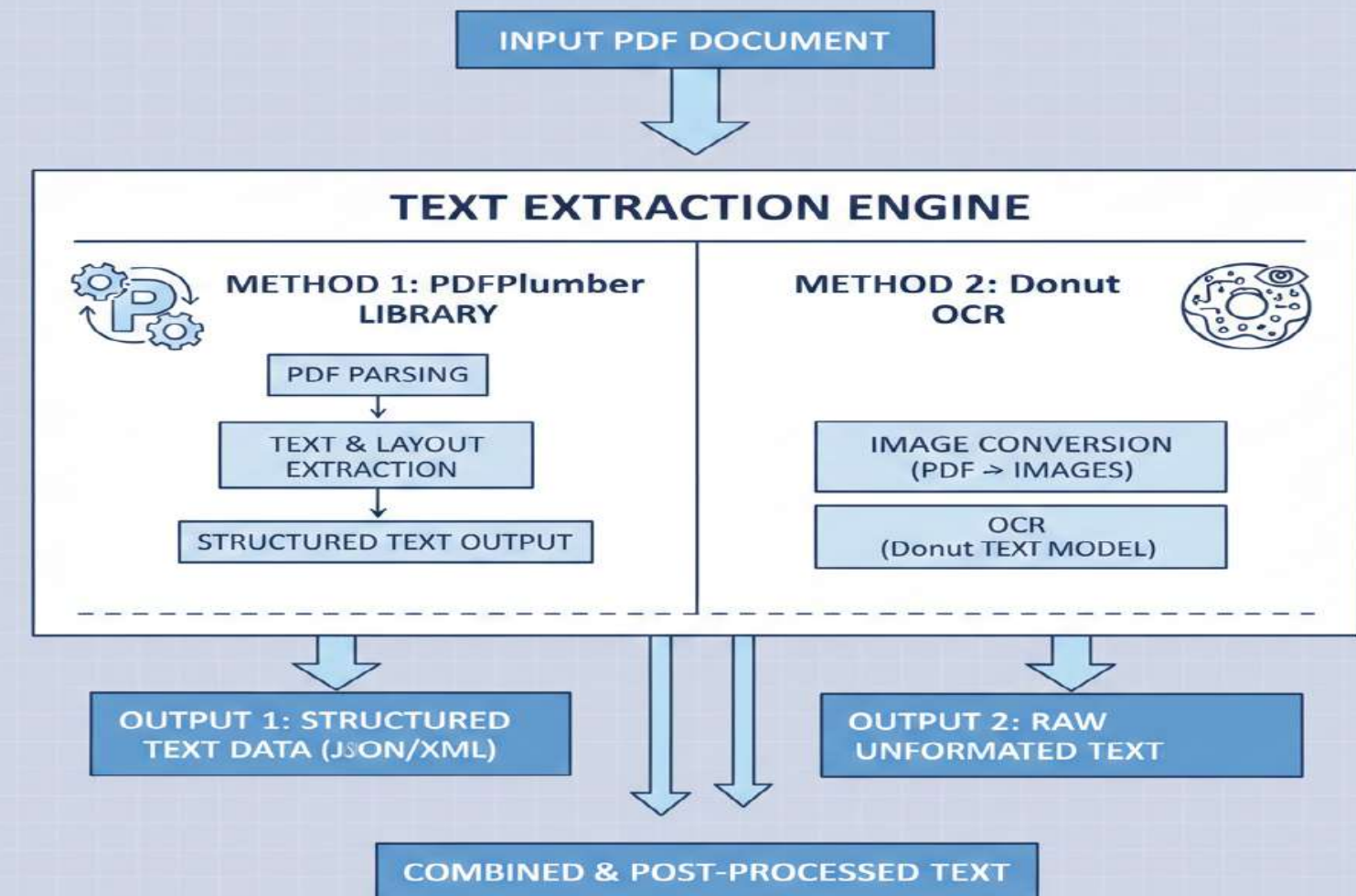
PDFPlumber

- Python library
- Extracts only selectable text from PDF
- High Accuracy and no context limit



DONUT OCR

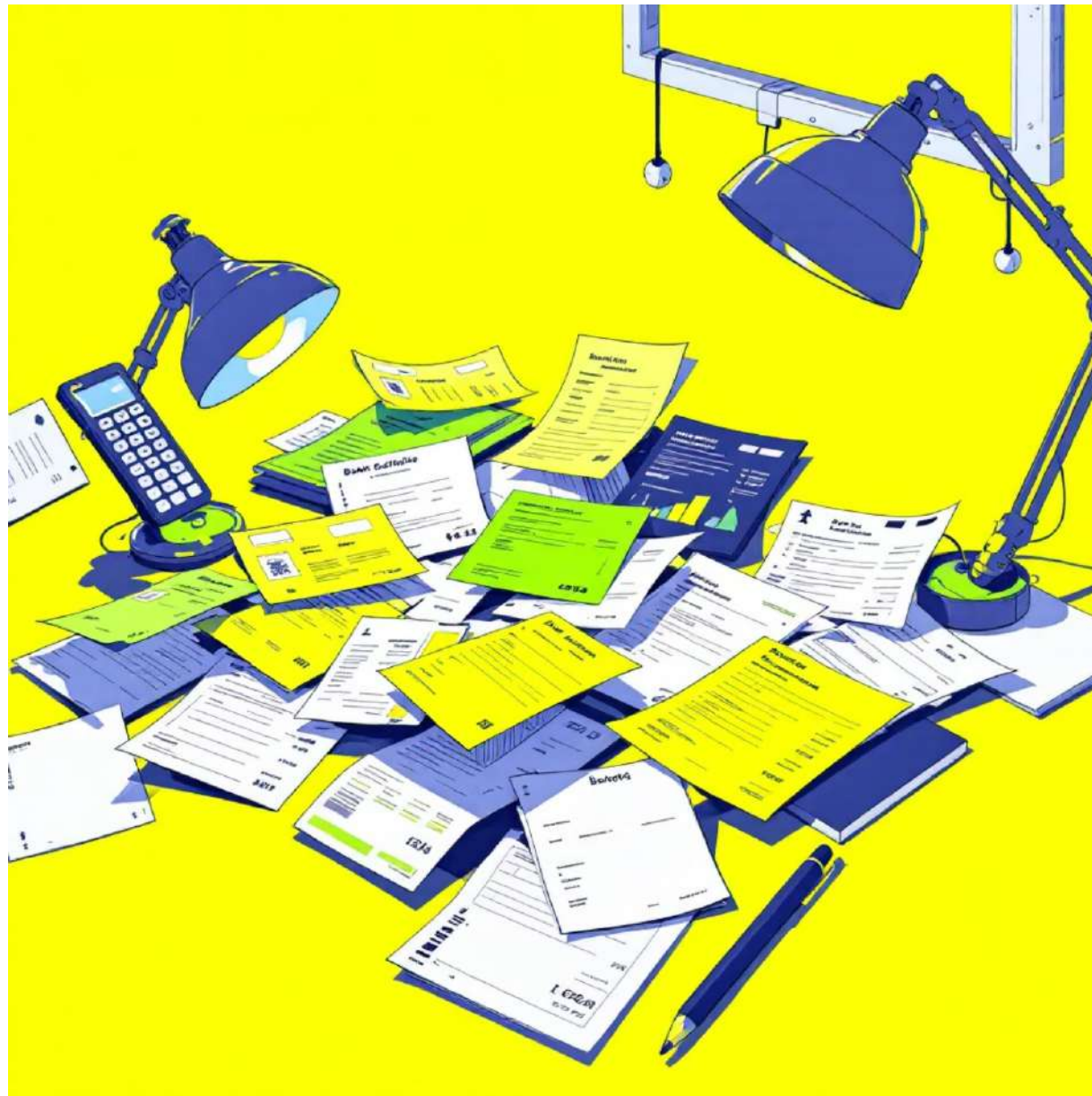
- Used a pre-trained Model from hugging face
- Converts each page to an image and then extracts texts
- Can extract text from scanned documents
- Low accuracy
- Limited by the context window



Why Dual Pipelines Matter

The Challenge

Bank statements are often provided in the scanned form or the next in the pdf file is not always selectable, that's where the libraries like PDFplumber fails



The Solution

Use a OCR-free Vision Transformers like DONUT in parallel to tackle challenging scanned documents. Gemini 2.0 reasoning structures and categorizes all output.



Modern Stack for Performance

Streamlit

We used **Streamlit** to create interactive dashboards for visualisation

Donut

Donut: provides OCR-free Transformer-based document understanding

Gemini 2.0 Flash

Enforces JSON schema for precise data extraction

PyTorch & CUDA

GPU-accelerated tensor computation power

Pdfplumber & Pdf2image

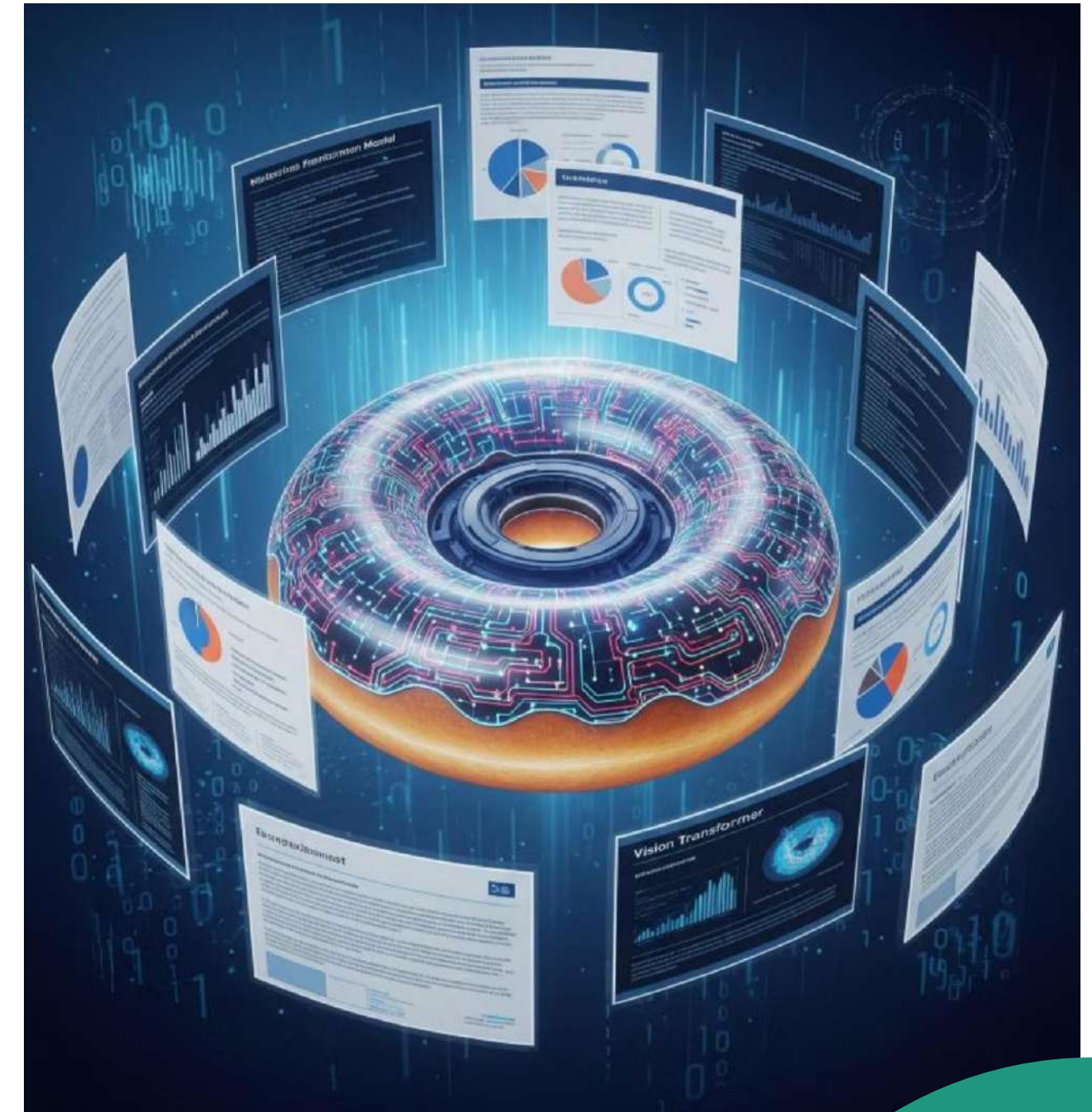
to extract text and tables from PDFs & to perform rasterization for processing



Interactive Dashboard & Visualisation

Streamlit creates highly interactive dashboards that visualize processed financial data clearly and intuitively. The frontend interface provides easy access to key metrics, allowing users to explore and analyze data quickly.

- Real-time data exploration
- Intuitive metric visualization
- Fast analytical insights



Vision Model Donut



Vision Transformer Architecture

Donut performs OCR-free document understanding using advanced Vision Transformer models.



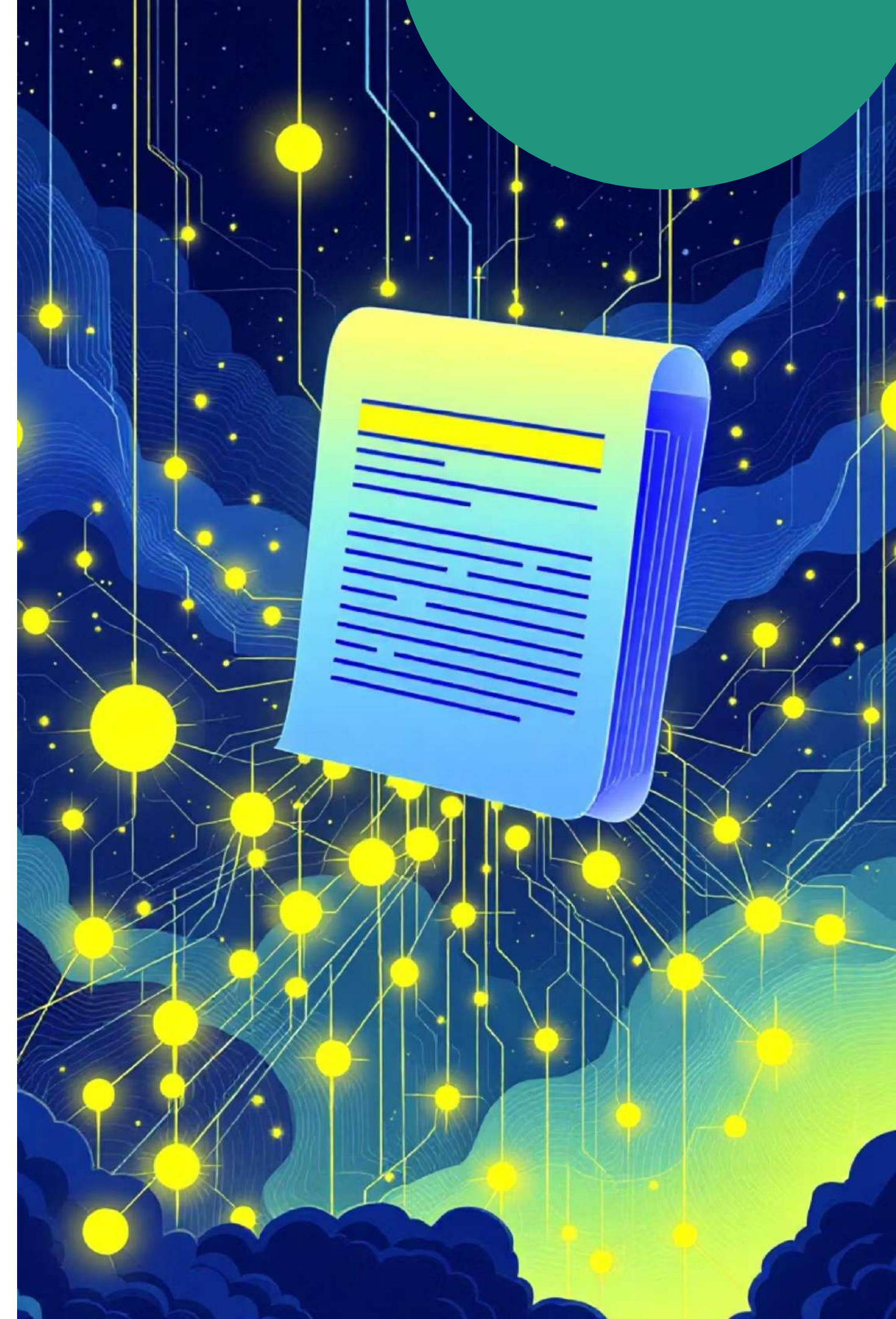
Proven Technology

Developed by the naver-clova-ix team for production-grade document analysis.



Flexible Layout Handling

Efficiently processes various document layouts without traditional OCR preprocessing.



Gemini 2.0 Flash: LLM Reasoning

Semantic Extraction

Performs powerful semantic data extraction from complex documents

Schema Enforcement

Ensures strict JSON schema compliance for consistent output

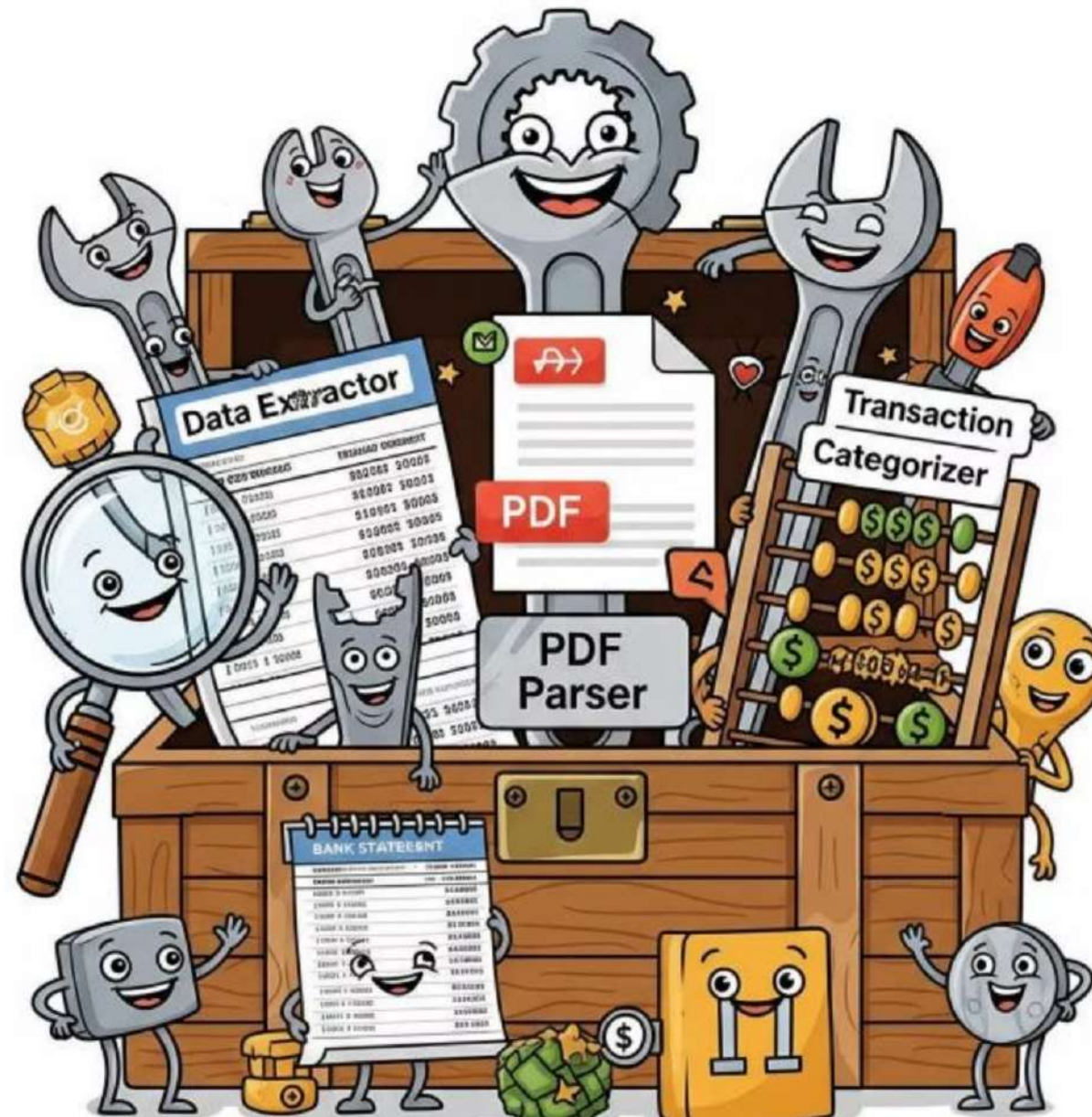
Unified Structure

Structures and categorizes all extracted data systematically



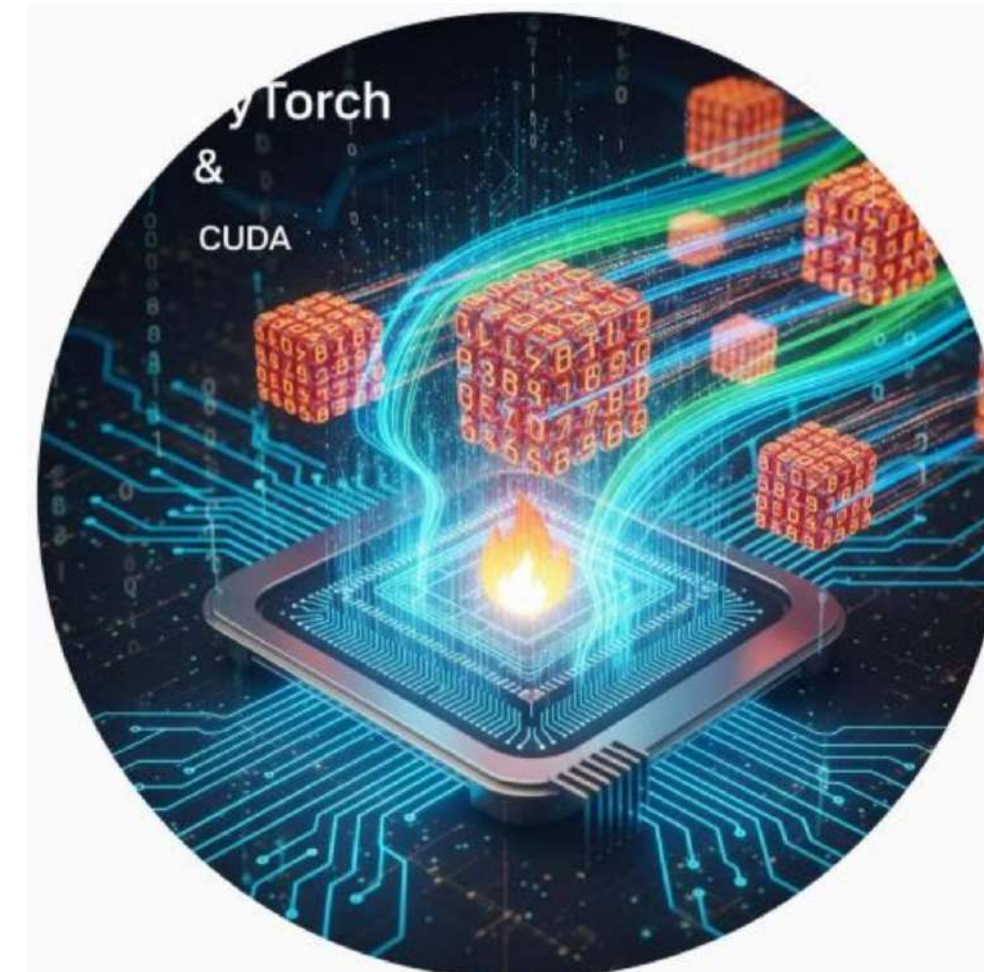
PDF Processing & Computation

Document Processing



pdfplumber extracts text and tables from PDFs while **pdf2image** performs rasterization for visual processing.

GPU Acceleration

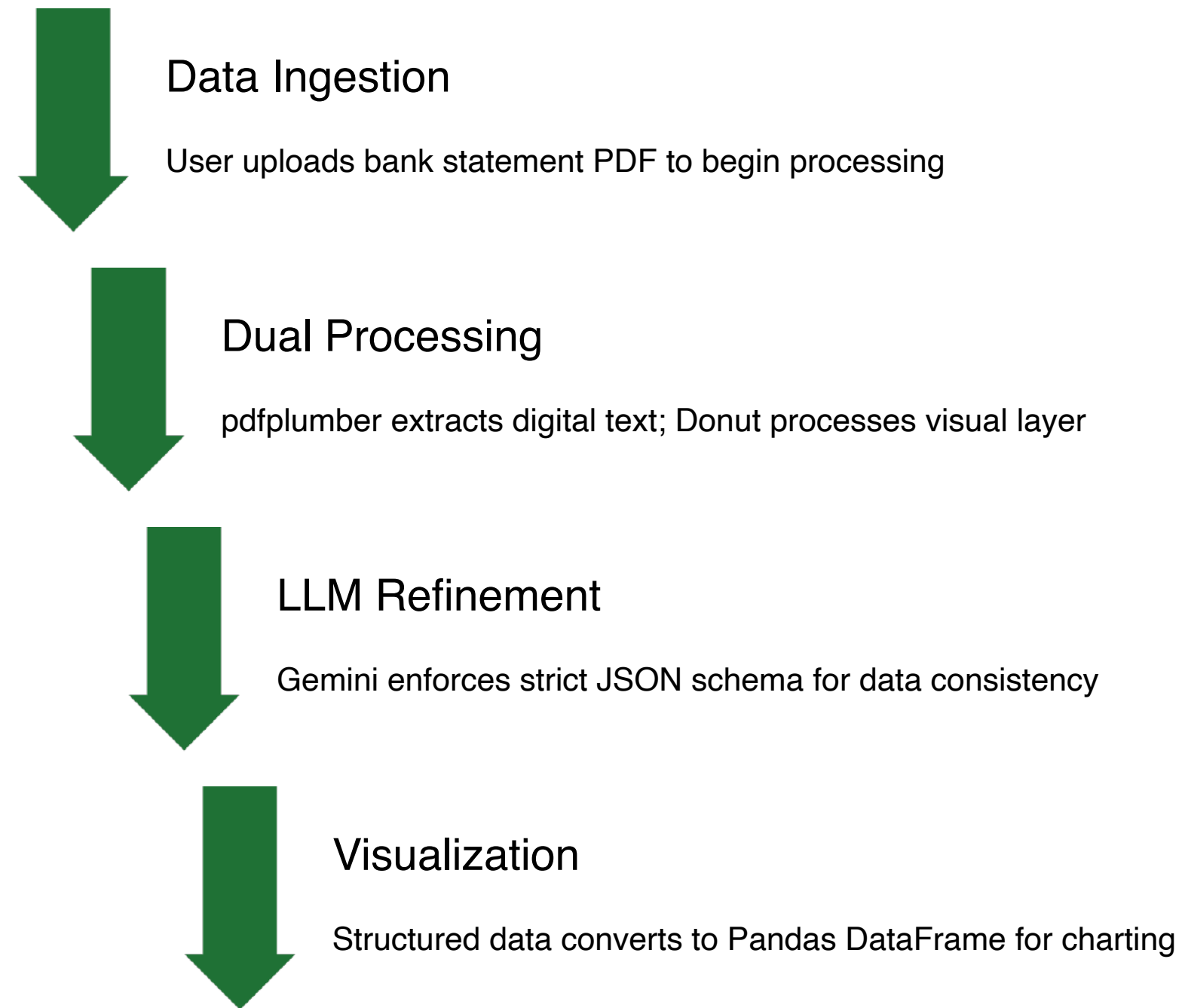


PyTorch enables tensor computation while **CUDA** provides GPU-accelerated processing for high-speed data analysis.



SYSTEM FLOW

Complete Architecture Flow



Text Extraction

PDFPlumber

Why we use it:

- **Precision:** Unlike standard OCR, it extracts text by identifying precise x, y coordinates and character metadata within the PDF stream.
- **Speed:** It is computationally lightweight compared to computer vision models, making it ideal for the bulk of the 100s pages.
- **Accuracy:** It avoids "hallucinations" common in visual text recognition by reading the embedded text layer directly.

How we use it:

- **Sharding:** The document is split into small "Auditor Chunks" (4–8 pages) to stay within the LLM's optimal attention span.
- **Metadata Recovery:** It captures font styles and positions to help distinguish between headers and transaction rows.

Text Extraction

DONUT (OCR)

Why we use it:

- **OCR-Free Parsing:** DONUT (Document Understanding Transformer) is a visual model that reads the page as an image, making it the fallback for "dead" PDFs (scanned images) where no text layer exists.
- **End-to-End:** It maps pixels directly to structured information without requiring a separate OCR engine like Tesseract.

How we use it:

- **Preprocessing:** We use **PIL** and **ImageChops** to trim whitespace and normalize DPI to 150, ensuring the model isn't overwhelmed by high-resolution noise.
- **Vision-Encoder-Decoder:** The model encodes the image and decodes it into a string of tokens representing the financial text.

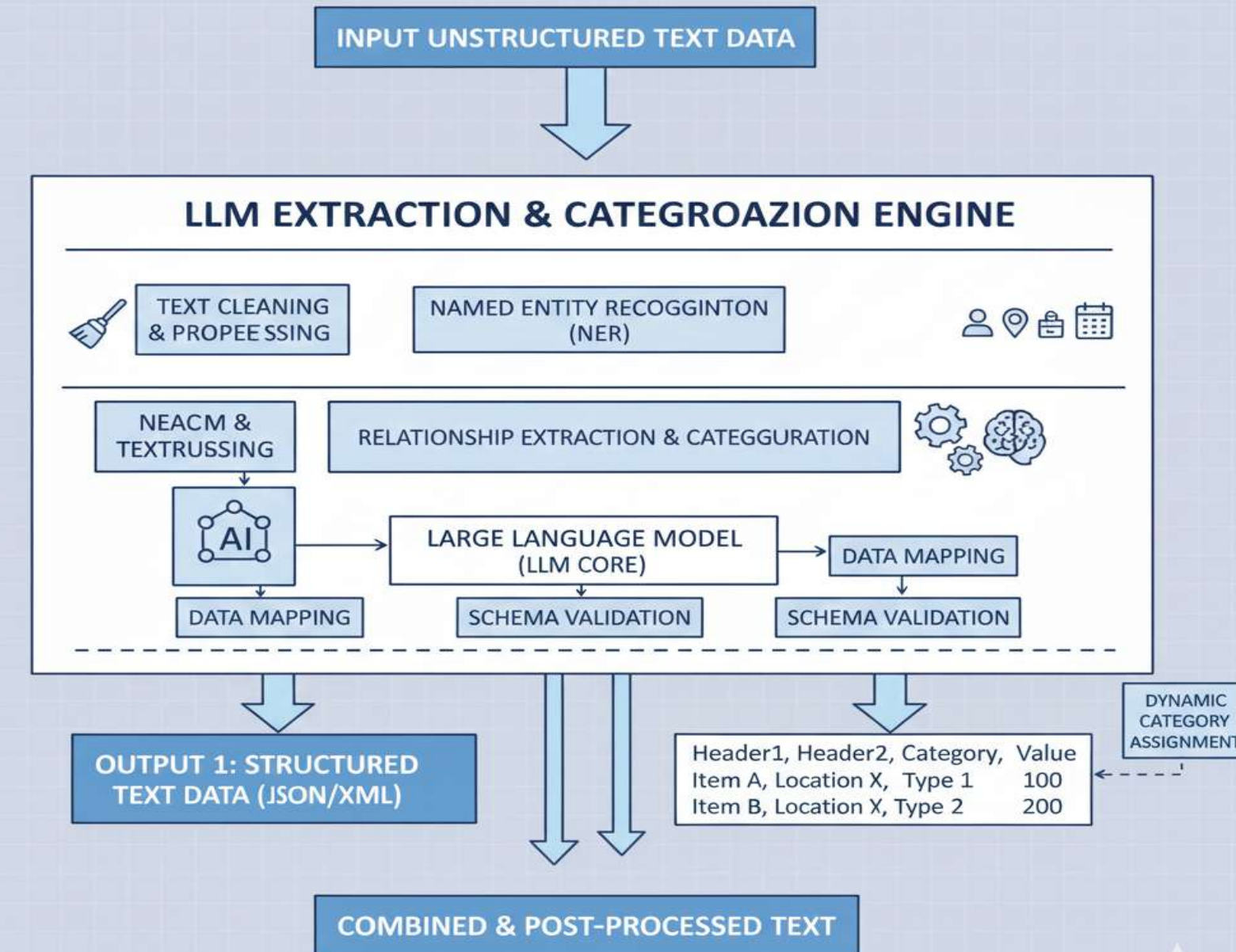
The LLM Engine – Why and How?

Why we use it:

- **Schema Enforcement:** We use Gemini to transform messy, unstructured strings into valid, machine-readable JSON.
- **Reasoning:** Unlike a regex script, the LLM can "read" a description and understand that "PSO Al Mansoor" refers to a fuel station.

How we use it:

- **Strict Schemas:** We define a JSON schema structure that forces the model to output specific keys: date, description, debit, credit, and balance.
- **Temperature Control:** We set **temperature=0.0** to ensure deterministic, consistent results—preventing the AI from "getting creative" with financial numbers.



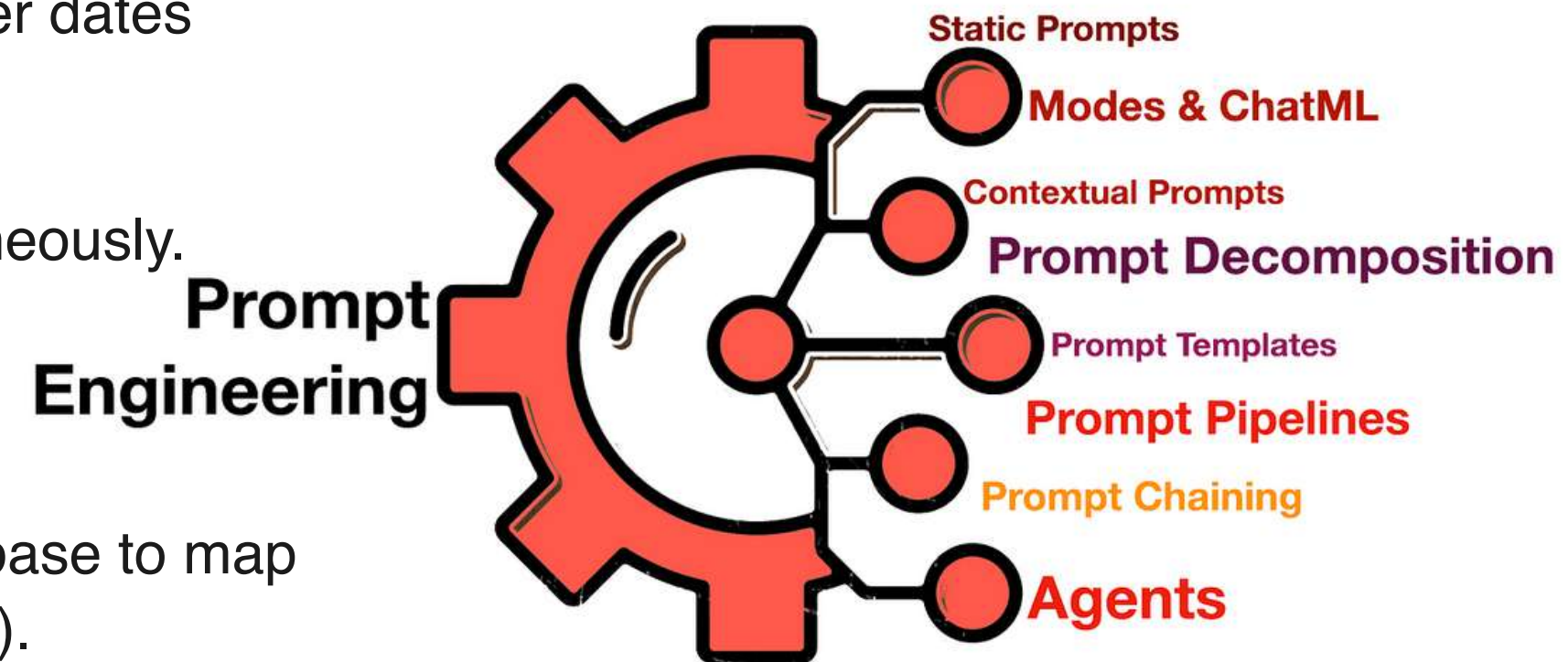
Prompt Engineering & Intelligent Categorization

Logic behind the prompt:

- **Context Persistence:** We explicitly instruct the LLM to carry over dates from previous rows if a row is missing its date stamp.
- **Mutually Exclusive Logic:** The prompt enforces a rule that a transaction cannot have a value in both **debit** and **credit** simultaneously.

Categorization Strategy:

- **Merchant Inference:** The LLM uses a predefined knowledge base to map keywords to categories (e.g., "KFC" --> Dining, "Shell" --> Fuel).
- **Transfer Isolation:** It identifies keywords like "Zelle," "Raast," or "Zaira" to flag peer-to-peer money movement.



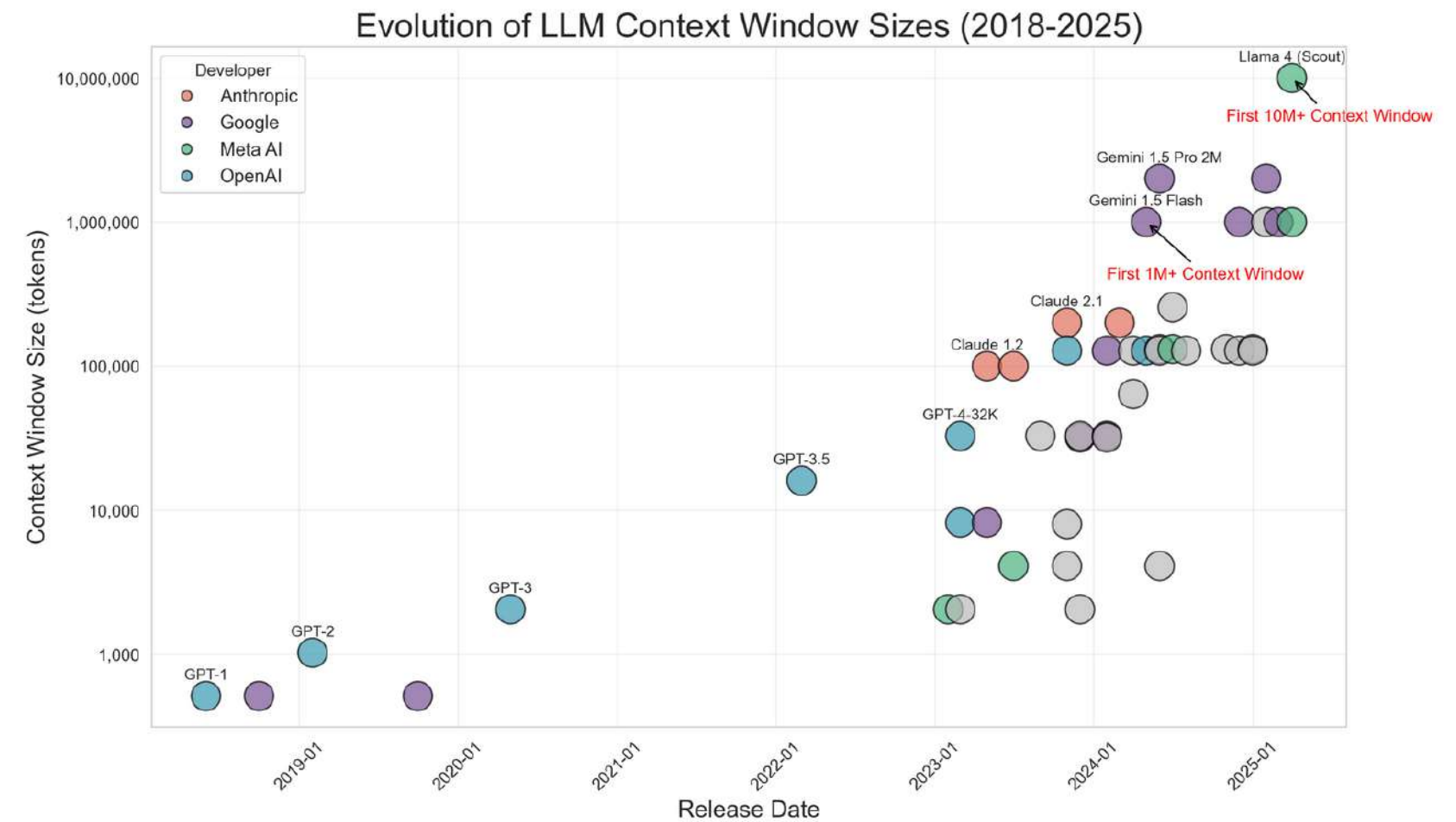
Handling "Token Limits" via Chunking

Problem

A 100s-page bank statement is too large for a single LLM prompt and can exceed the context window.

Solution

- We split the PDF into 5-10 page 'chunks'.
- We send each chunk to Gemini separately to respect output token limits.
- We stitch the JSON results back together into a single master list.



Data Privacy & Security (Critical for Fintech)

Policy

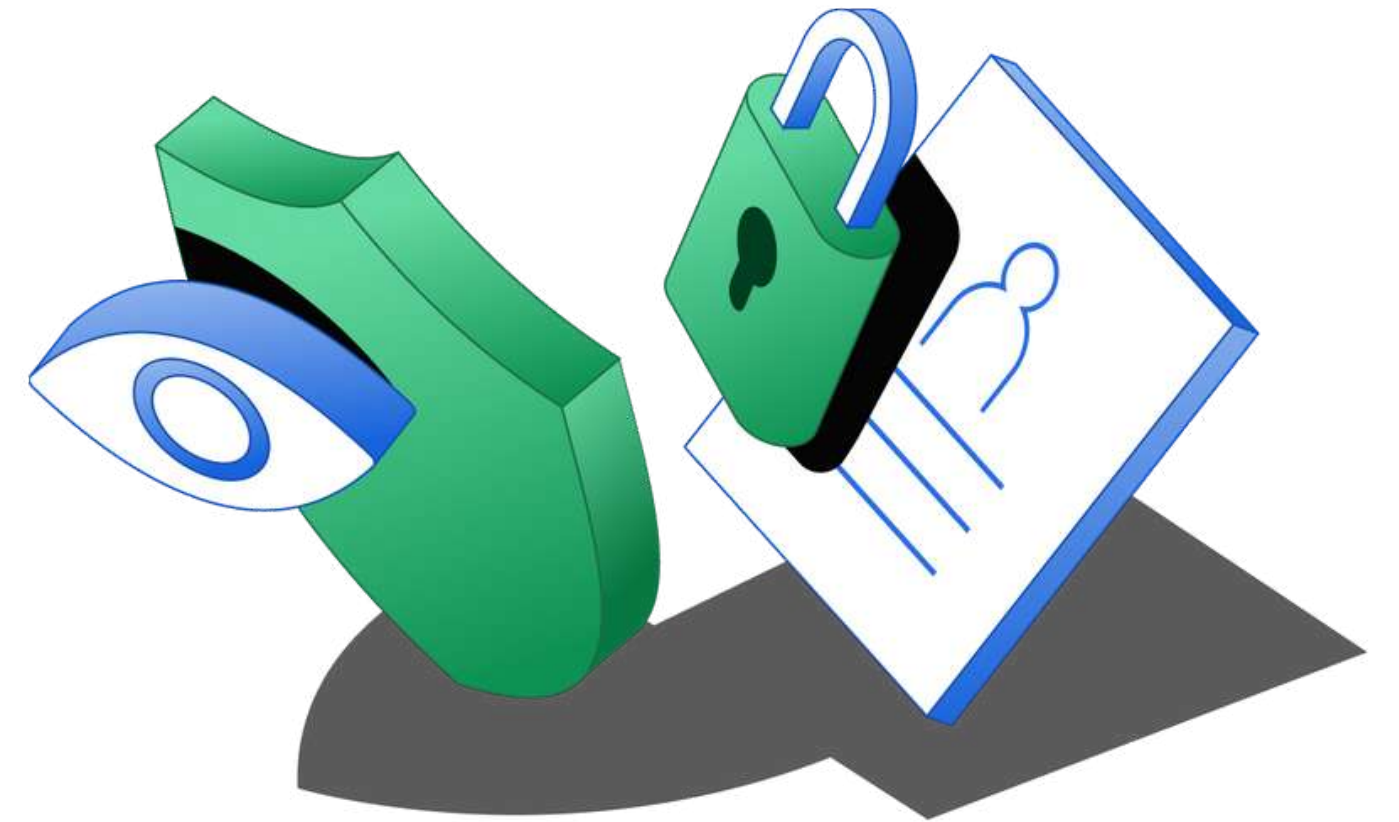
Handling financial data requires security.

Our Implementation

```
os.path.exists(temp_path): os.remove(temp_path)
```

User data exists on the server only during processing and is wiped immediately after.

Why: For a financial tool, security is a technical feature, not just a policy.



Pipeline Comparison

$$Score = \max \left(0, \left(1 - \frac{|Count_{ref} - Count_{cand}|}{Count_{ref}} \right) \right) \times 100$$

Similarity : Ranges between **85%-95%** PDFPlumber as a benchmark for DONUT

The screenshot shows the 'Financial Statement Analyzer' web application. The interface is dark-themed with a sidebar on the left and a main content area on the right. The sidebar contains an 'Upload Document' section with a file upload area (showing 'FullStmt_17244246046... 27.9KB') and a 'Pipeline Settings' section with a checkbox for 'Run Pipeline 2 (Donut OCR)' and a 'Process Statement' button. The main content area features the application title 'Financial Statement Analyzer' and a description 'Extract, Compare, and Visualize your Bank Statement data using Hybrid AI (OCR + LLM)'. Below this, there are two tabs: 'Analytics Dashboard' and 'Extraction Details'. The 'Extraction Details' tab is active, showing a comparison between 'Pipeline 1 (PDFPlumber)' and 'Pipeline 2 (Donut OCR)'. Pipeline 1 has 'Extracted 46 transactions' and Pipeline 2 has 'Extracted 40 transactions'. A 'Similarity Score (vs Pipeline 1)' of '86.96%' is displayed. There are buttons for 'Download JSON', 'Download CSV', and 'View Raw JSON' for both pipelines.

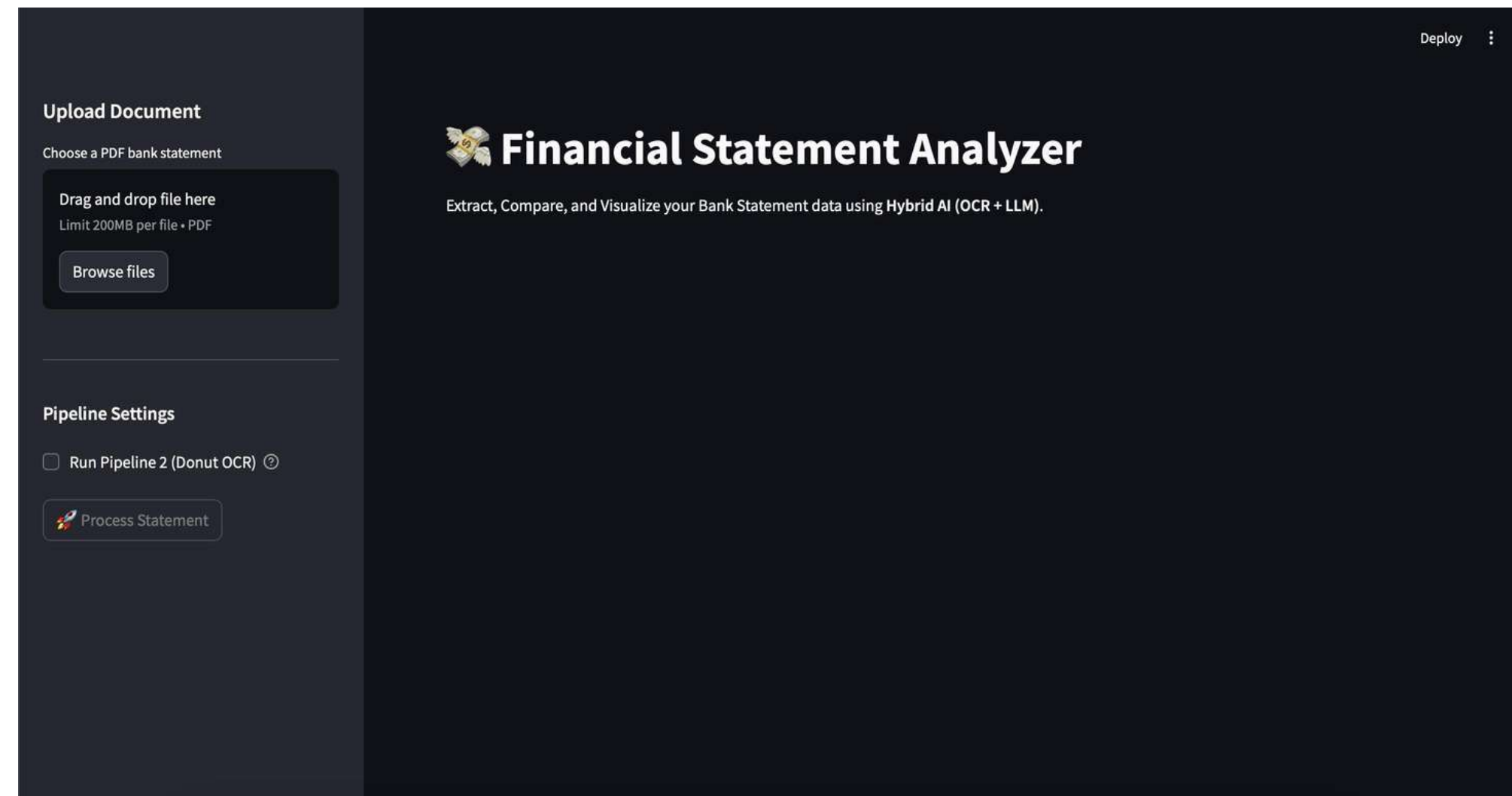
The Dashboard Interface

Framework: Built using Streamlit for rapid, responsive deployment.

User-Centric Design: Simple "Drag & Drop" functionality for non-technical users.

Hybrid Control: Users can toggle between Pipeline 1 (Text-based) for speed or Pipeline 2 (Donut Vision) for scanned images.

Security: Data is processed in-memory and not stored permanently.



Instant Financial Health Check

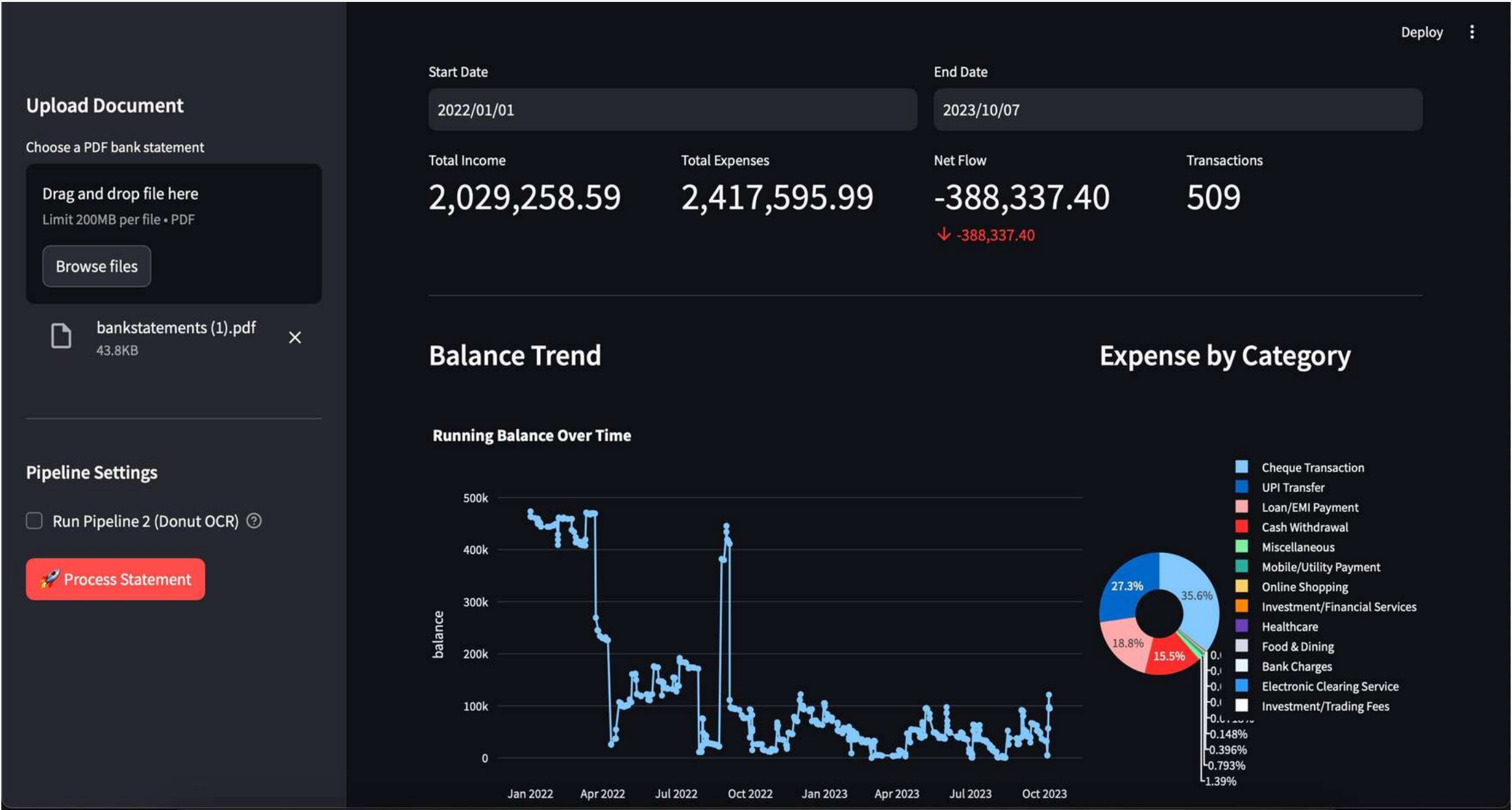
1. **High-Volume Processing:** Successfully parsed 509 transactions from 2022 - 2023.

2. Automated Risk Detection:

- Income: ₹2.02 Million
- Expenses: ₹2.41 Million
- Net Flow: -₹388,337 (Deficit Detected).

3. **Volatility Analysis:** The "Balance Trend" chart identifies critical events, such as the sharp 50% liquidity drop observed in October 2022.

4. **Visualizing Habits:** The Pie Chart reveals that 35.6% of activity is "Cheque Transactions".

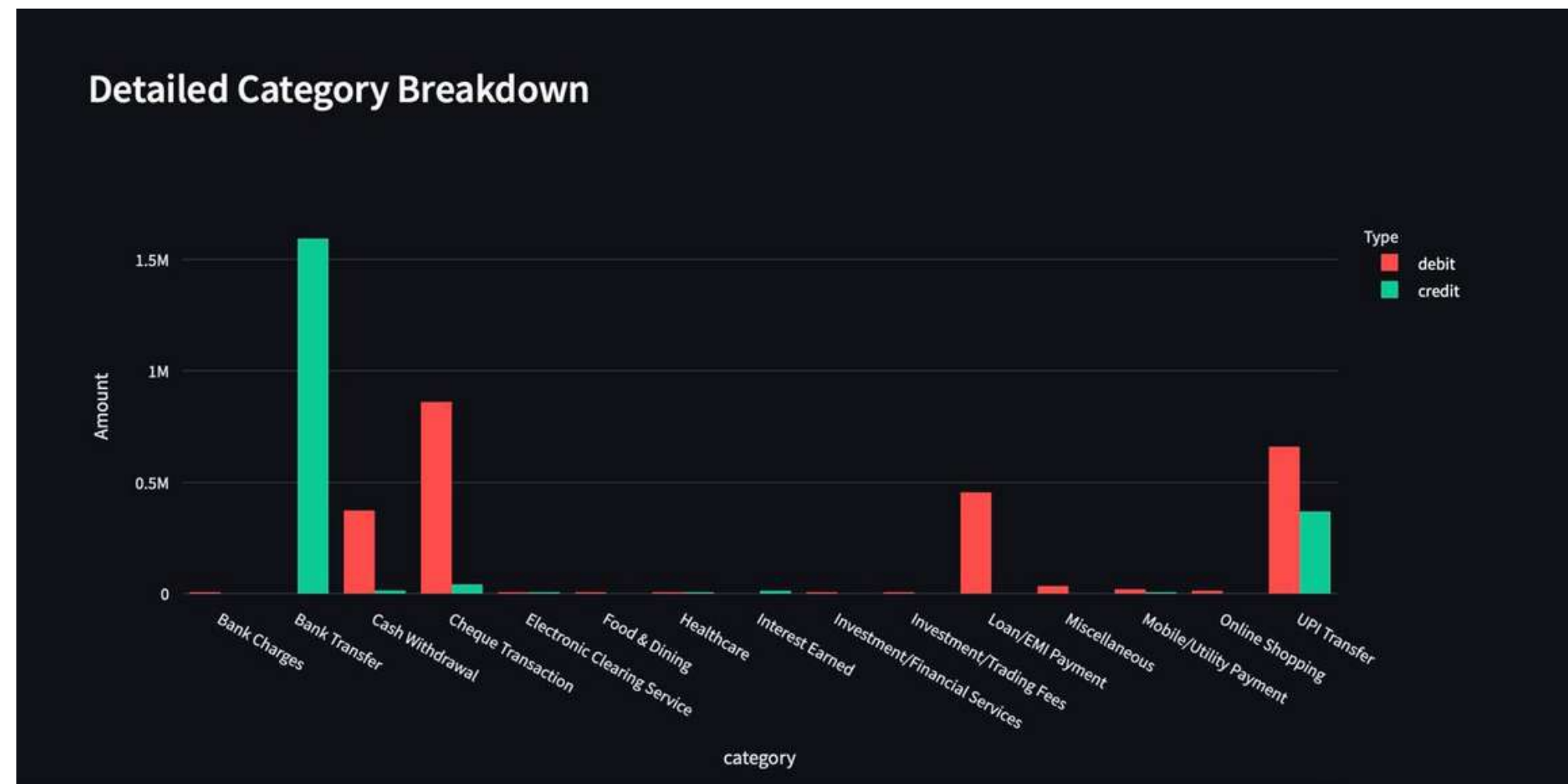


Automated Categorization

- **LLM Reasoning:** Google Gemini classifies ambiguous descriptions (e.g., "UPI-SWIGGY" → "Food & Dining").

- **Spending Profile:**

- Highest Credit (Green): Bank Transfers.
- Highest Debit (Red): Cheque Transaction and UPI Transfer.



Granular Transaction Log

- **Full Transparency:** Every extracted row is visible for manual verification.
- **Smart Description:**
 - Row 507: Identified "UPI HDFCBANK" as "Loan/EMI Payment".
 - Row 505: Identified "ECS" as "Electronic Clearing Service".
- **Exportable:** Data can be downloaded as CSV/JSON for ERP integration.

Transaction Log

	date	description	category	debit	credit	balance	amount
508	2023-10-07 00:00:00	UPI LAGADEKSA	UPI Transfer	100	0	94,955.47	-100
507	2023-10-06 00:00:00	UPI HDFCBANK	Loan/EMI Payment	26,286	0	95,055.47	-26,286
506	2023-10-06 00:00:00	UPI DADAKHALA	UPI Transfer	0	23,586	121,341.47	23,586
505	2023-10-06 00:00:00	ECS	Electronic Clearing Service	0	107.8	97,755.47	107.8
504	2023-10-06 00:00:00	CHEQUE	Cheque Transaction	0	41,050	97,647.67	41,050
503	2023-10-05 00:00:00	UPI AYUBRAJE	UPI Transfer	48	0	56,597.67	-48
502	2023-10-05 00:00:00	NEFT	Bank Transfer	0	51,905	56,645.67	51,905
500	2023-10-04 00:00:00	UPI DADAKHALA	UPI Transfer	220	0	5,026.67	-220
499	2023-10-04 00:00:00	UPI DADAKHALA	UPI Transfer	26,286	0	5,246.67	-26,286
501	2023-10-04 00:00:00	UPI DADAKHALA	UPI Transfer	286	0	4,740.67	-286



The Limitations of

A **Limited to bank statement only**

This current project is only limited to the scope of bank statement

B **Low accuracy for scanned documents**

OCR can be the only pipeline if it is made robust and more accurate

C **Read tables & check-boxes**

Current model needs fine tuning to be able to handle all kinds of check-boxes and tabular data often find in forms.

Future Vision: End-to-End Document Understanding with DONUT

Our strategy is to harness the power of Donut (Document Understanding Transformer) to achieve seamless, end-to-end document understanding.

- **Input:** Raw document image.
- **Output:** Structured JSON data, completely bypassing traditional OCR steps.
- This approach can be our single go-to approach for handling all kinds of documents if it is made robust and accurate.



Fine-Tuning DONUT for General Document Types



Receipts

Extracting purchase details and vendor information.



Invoices

Automating data capture for billing and accounts payable.



Forms

Processing structured and semi-structured forms efficiently.



Reports

Parsing complex data from financial and operational reports.



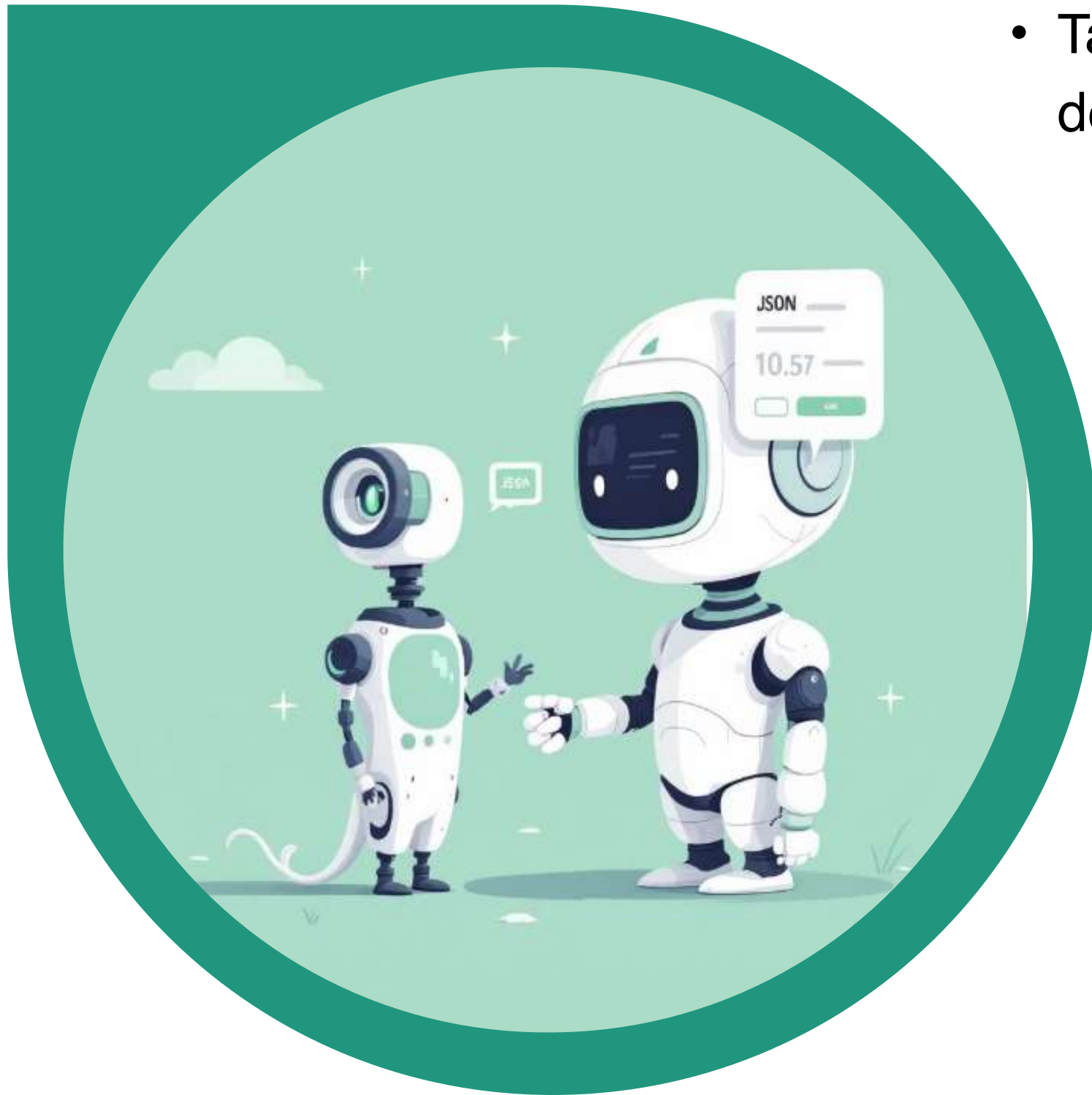
TRAINING INPUTS & EXPECTED BENEFITS

Training Inputs

- Document Image: The raw visual input of the document.
- Target JSON Schema: Predefined labels and structure for the desired output.

Expected Benefits

- Improved Accuracy
Enhanced field extraction precision.
- Reduced Dependency
Less reliance on fragile OCR heuristics.
- Better Robustness
performance across diverse layout variations.



DATA STRATEGY FOR GENERALIZATION

Dataset Expansion



- Multi-domain documents
- Synthetic document generation
- Layout diversity (tables, paragraphs, forms)

Annotation Strategy



- Schema on the go.
- Semi-automated labeling processes

Scalability



- Rapid integration of new document types with minimal retraining effort.
- Providing API solutions to business so they can integrate our solution

TECHNICAL ENHANCEMENTS FOR PEAK PERFORMANCE

Model Improvements

Larger Backbone

Swin-based encoders for enhanced feature extraction.

Multi-Language

Fine-tuning for global applicability.

Low-Quality Scans

Improved handling of imperfect document images.

Training Enhancements

Curriculum Learning

Progressing from simple to complex layouts.

Data Augmentation

Rotation, blur, and noise for robustness.

Mixed-Precision

Optimized training for performance gains.

FUTURE DIRECTIONS: INTEGRATING DONUT INTO THE ENTERPRISE

LLM-ASSISTED
VALIDATION AND
REASONING
CAPABILITIES.

REAL-TIME
INSTANTANEOUS
DOCUMENT
PROCESSING.

ERP/SAP
INTEGRATION
SEAMLESS DATA
FLOW INTO CORE
BUSINESS SYSTEMS.

OUR ULTIMATE GOAL IS TO EMBED DONUT AT THE HEART OF ENTERPRISE OPERATIONS, DRIVING
EFFICIENCY AND INTELLIGENCE.

Summary



The Challenge:

- Financial audits are currently bottlenecked by manual data entry.
- Processing complex, multi-page bank statements is slow, unscalable, and error-prone.



Our Solution:

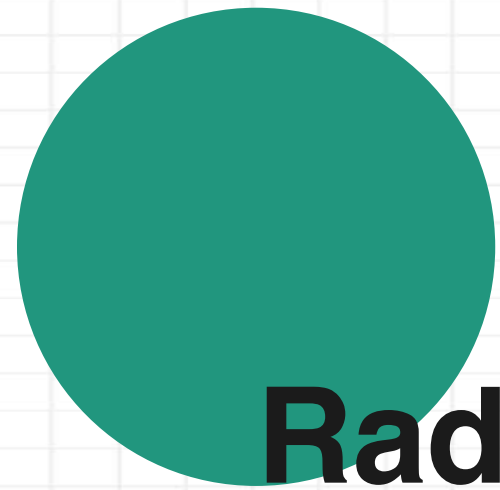
- We developed an end-to-end Hybrid AI Pipeline.
- Combined Computer Vision (Donut) for scanning with LLMs (Google Gemini) for reasoning.



The Outcome:

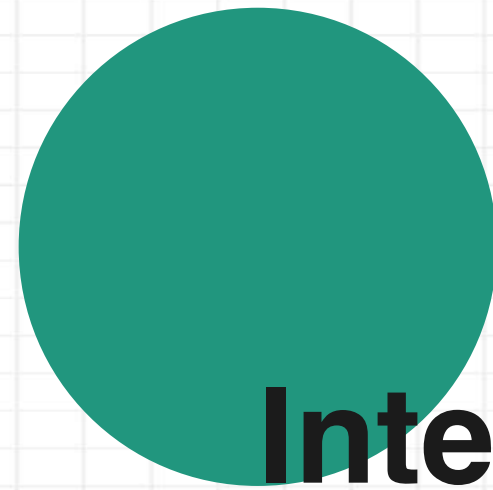
- A fully functional dashboard that transforms raw, unstructured PDFs into 100% digital, analyzed financial data in seconds.

Key Takeaways



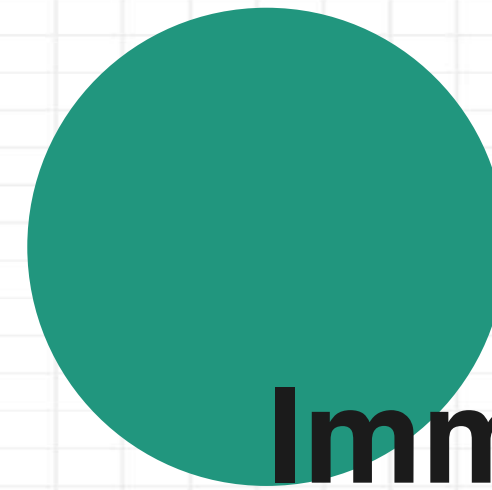
Radical Efficiency:

- Reduced document processing time from 45 minutes to under 1 minute per file.
- Eliminated manual data entry errors completely.



Intelligent Context:

- The system goes beyond OCR; it understands context.
- Successfully categorizes ambiguous transactions without manual rules.



Immediate Value:

- Provides instant visibility into financial health.
- Automatically flagged critical risks, such as the - ₹388k deficit, enabling faster decision-making.

Thank you Everyone



**Any
questions**

